Gogulnath Sathish Amuthavalli

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Professional Summary

MSc Robotics student at the University of Bristol specializing in Sensor fusion, ROS2, and Control theory for autonomous systems. Skilled in designing real-time perception, localization, SLAM and control pipelines. Strong interest in Vision-Language-Action (VLA) and AI-driven robotic autonomy, and seeking a role in robotics research focused on intelligent, perception-driven robotic systems.

Passionate about building reliable intelligent robots for real-world applications.

Education

- **MSc Robotics**
  University of Bristol, UK
  Sep 2025 – Sep 2026 (Expected)
- **Bachelor of Engineering in Mechatronics**
  Sastra University, India
  Graduated Augest 2024
  CGPA: 7.02 / 10

Project

Implementing VLA in the HITL – Ongoing

- Focusing on the pipeline architecture in VLA.
- Understanding the current challenges in implementing VLA in the robot.
- Trained ACT Policy with a custom task
- Exploring State-Space-Systems(mamba) and Robomamba for optimizing latency

Stability Enhancement for Leader-Follower by Using Bayesian Filter

- Developed a probabilistic framework to achieve reliable directional stability in an IR Sensor-based leader follower robotic system.
- Focused on making the Leader-Follower robust with Bayesian Filter.

Robust Apple Detection in Orchard Environment Using Machine Vision and SVM

- Built an image processing pipeline with stages using Support Vector Machine for detecting and counting the apples in Orchard environment.
- Tested mainly on machine-learning method to obtain accurate results.

Indoor Drone Localization For Mobile Cable Driven Parallel Manipulators

- The central aim of this project is to autonomously localize drones within indoor environments, particularly for mobile cable-driven manipulators.
- ArUco Markers are utilized to achieve drone localization indoors, facilitating their performance of necessary tasks.
- The methodology begins with drone construction, integrating essential components, and testing drone localization with markers in simulated environments to analyse results.
- Cameras are calibrated for detection purposes. The drone underwent testing in various modes like Guided, Guided-No-GPS, and Stabilize to comprehend their functionalities.

Semi-Autonomous Shopping Cart

- Worked in SITL for tracking human movement in department stores.
- Focused on the Sensor integration.

Certifications

- Self Driving and ROS-Learn by Doing. Odometry and Control
- Advanced Kalman Filtering and Sensor Fusion
- IELTS Academic Band 6.5

Skills

Analytical & Strategic: Mathematical and algorithmic reasoning • Hypothesis-driven experimentation • Statistical evaluation of perception and learning models • Optimization and control analysis • Interpreting real-world robotic data under noise and uncertainty • Translating research insights into deployable robotic solutions

Engineering & Collaboration: Technical documentation for robotics systems • Coordination across hardware, software workflows • Structured presentation of technical results and experiments • strong ability to analyze complex robotic perception and control systems

Communication: Technical documentation | Clear written and verbal communication | IELTS Academic Band 6.5 (B2)